

JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Permutations And Combinations - Core Application	14 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Permutations And Combinations - Core Application (High Yield — 14 PYQs)

Core Principle: Key problem pattern in Kcet Permutations And Combinations.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Stateme..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of four digit even number that can be formed using the digits 0,1,2 and 3 without repetition is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?

- (A) 720
- (B) 360
- (C) 120
- (D) 180

Correct Answer: None

Solution Analysis: The characters in "KCET 2025" are: K,C,E,T,2,0,2,5 There are 8 characters in total, and the digit 2 is repeated twice. We need arrangements that start with K and end with 5 . So fix K in the first position and 5 in the last position: K_____5 Now we have to arrange the remaining 6 characters: C,E,T,2,0,2 If all were different, the number of arrangements would be: $6!$ But the digit 2 appears twice, so we divide by $2!$: $(6! / 2!) = (720 / 2) = 360$ So the number of ways is: 360 Correct option: Option B .